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Talib Mirza & Preeti Bhakta

# Build A **Clap-** And **Gesture-Controlled Robot**

Sani Theo

This is a two-in-one project to control a robot in four directions (forwards, backwards, right and left) either by clapping or through simple gestures. It basically explains how a robot can be controlled using sound or gesture.

A switch is used to change mode of operation from sound to gesture control. You can make the robot to follow your yoga or dance moves, or react to various environmental sounds.

The robot control circuit has separate sections for clap and gesture. Both the sections have their own sensors. In clap section, an electret mic works as the sound sensor; in gesture mode, an accelerometer is used as the sensor.

Through the sound section you get clap-based switching control using a sound sensor, timer and decade counter. The gesture section uses a motion sensor like an accelerometer, Arduino-based ATmega328P microcontroller (MCU), radio frequency (RF)-based wireless control and encoder.

The entire robot is divided into two control sections: clap and gesture. Clap control section has only receiver while gesture control section has transmitter and receiver.

The circuit diagram of the clap- and gesture-controlled robot is shown in Fig. 1. For clarity, clap and gesture circuits are explained in separate sections.

## Clap control

The clap control circuit is built around electret condenser microphone (MIC1), op-amp NE5534 (IC1), preset (VR1), timer NE555 (IC2), decade counter 4017B (IC3), L293D motor driver (IC4), two DC motors (M1 and M2) and a few other components.

Each clap sound works as a command signal to move the motor. The robot will move in different directions with four consecutive claps, namely, forwards, backwards, left and right, while the fifth clap stops the robot. VR1 preset is used to control sensitivity of the sensor, which, in turn, decides the range (distance) of control.

IC1 is used as the audio sound amplifier to amplify clapping sound received from the mic. IC2 generates time delay and provides it to clock input of counter IC3. That is, output from pin 3 of IC2 goes to pin 14 of IC3, providing a clock pulse when you clap in front of the mic.

For the first clap, Q0 output of IC3 will be high, turning LED2 on. For the next clap, Q1 will be high, turning LED3 on. Simultaneously, Q0 will go low and LED2 will turn off.

Similarly, Q2 and Q3 will be high for the third and fourth claps, respectively. Clap sequence and LED status are shown in Table 1.

Output from IC3 also goes to motor driver IC4 via diodes D2 through D9 to move the robot. When Q0 is high, the motor moves in forward direction. When Q1 is high, it moves in backward direction. When Q2 is high, the motor moves in right direction, and when Q3 is high, it moves in left direction. The fifth clap resets IC3, and Q0 through Q3 outputs along with all corresponding LEDs switch off, and the robot stops.

For each clap, the respective LED (LED2 through LED5) glows to indicate direction of robot's movement. Direction of motors can be changed by changing polarity of both the motors (M1 and M2).

**PARTS LIST**

| | |
|---|---|
| *Semiconductors:* | |
| IC1 | - NE5534/ LM741 op-amp |
| IC2 | - NE555 timer |
| IC3 | - 4017B decade counter |
| IC4 | - L293D motor driver |
| IC5, IC8 | - 7805, 5V regulator |
| IC6 | - 7806, 6V regulator |
| IC7 | - HT12D decoder |
| IC9 | - ATmega328 MCU |
| IC10 | - LM1117-3.3, 3.3V regulator |
| IC11 | - HT12E encoder |
| D1 | - 1N4007 rectifier diode |
| D2-D13 | - 1N4148 signal diode |
| LED1-LED9 | - 5mm LED |
| *Resistors (all 1/4-watt, ±5% carbon):* | |
| R1, R3-R5, R8, R18 | - 10-kilo-ohm |
| R2 | - 6.8-kilo-ohm |
| R6 | - 22-kilo-ohm |
| R7 | - 1-kilo-ohm |
| R9-R12 | - 680-ohm |
| R13 | - 47-kilo-ohm |
| R14-R17 | - 220-ohm |
| R19 | - 1-mega-ohm |
| R20 | - 750-kilo-ohm |
| VR1 | - 2-mega-ohm preset |
| *Capacitors:* | |
| C1, C2, C4, C7, C9 | - 0.1µF ceramic disk |
| C3 | - 10µF, 25V electrolytic |
| C5, C6 | - 100µF, 35V electrolytic |
| C8 | - 0.01µF ceramic disk |
| C10 | - 0.33µ ceramic |
| C11 | - 10µF, 16V electrolytic |
| C12, C13 | - 22pF ceramic disk |
| *Miscellaneous:* | |
| S1 | - DPDT switch |
| S2, S3 | - On/off switch |
| BATT.1 | - 12V battery (8x1.5V AA cells) |
| CON1 | - 2-pin terminal connector |
| CON2 | - 5-pin female Berg connector |
| MIC1 | - Condenser mic |
| XTAL1 | - 16MHz crystal oscillator |
| RF TX1 | - 433MHz RF transmitter module |
| RF RX1 | - 433MHz RF receiver module |
| ANT.1, ANT.2 | - 4cm long, single-strand wire for antenna |
| M1, M2 | - 5V/6V DC geared motor |
| | - 9V PP3 battery for CON1 |
| | - 12V battery holder |
| | - 18-pin IC socket (2 nos.) |
| | - 16-pin IC socket (2 nos.) |
| | - 8-pin IC sockets (2 nos.) |
| | - 20-pin IC socket for MCU |
| | - Robot chassis |
| | - Two wheels for M1 and M2 |
| | - Castor wheel for front wheel |
| | - PCBs |

*(All these are available from kitsnspares.com)*

**Table 1: Clap Sequence And LED Status**

| Clap | Output | LED | Direction |
|---|---|---|---|
| First | Q0 | LED2 on | Forward |
| Second | Q1 | LED3 on | Backward |
| Third | Q2 | LED4 on | Right |
| Fourth | Q3 | LED5 on | Left |
| Fifth | Q4 | LED2-LED5 off | Stop |